

LIANZHOU, China

WMO: 590720

Lat: 24.8108N

Lon: 112.3719E

Elev: 132

StdP: 99.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.6	(c) 2.9	(d) -4.8	(e) 2.6	(f) 8.9	(g) -2.6	(h) 3.1	(i) 6.9	(j) 6.5	(k) 9.9	(l) 5.9	(m) 9.9	(n) 1.5	(o) 0	(p) 0.308

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.1	(c) 36.1	(d) 26.0	(e) 35.1	(f) 25.9	(g) 34.1	(h) 25.7	(i) 27.4	(j) 32.6	(k) 27.0	(l) 32.2	(m) 26.6	(n) 31.5	(o) 2.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.2	(b) 22.0	(c) 29.1	(d) 25.8	(e) 21.4	(f) 28.7	(g) 25.5	(h) 21.0	(i) 28.4	(j) 88.0	(k) 32.7	(l) 86.2	(m) 32.4	(n) 84.5	(o) 31.7	(p) 29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.6	(b) 4.6	(c) 3.9	(e) -0.6	(f) 37.7	(g) 1.3	(h) 1.0	(i) -1.5	(j) 38.5	(k) -2.2	(l) 39.1	(m) -2.9	(n) 39.7	(o) -3.9	(p) 40.4		
(a) 5.6	(b) 4.6	(c) 3.9	(e) -1.4	(f) 28.2	(g) 1.4	(h) 0.6	(i) -2.3	(j) 28.6	(k) -3.2	(l) 28.9	(m) -3.9	(n) 29.2	(o) -4.9	(p) 29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	9.8	12.1	15.4	20.8	24.9	27.6	29.3	29.0	26.7	22.6	17.2	11.6
	DBStd	7.64	4.10	5.13	4.37	3.88	2.70	2.17	1.75	1.85	2.69	3.07	3.71	3.35
	HDD10.0	116	56	32	6	0	0	0	0	0	0	0	2	21
	HDD18.3	861	267	183	111	21	0	0	0	0	0	5	63	211
	CDD10.0	3993	48	91	175	325	462	528	599	588	501	389	217	69
	CDD18.3	1696	1	9	22	95	203	278	341	330	251	136	28	2
	CDH23.3	16478	4	52	108	569	1608	2848	4103	3746	2364	907	163	6
CDH26.7	6779	0	7	14	149	560	1130	1947	1697	966	279	29	1	
Wind	WSAvg	1.4	1.5	1.5	1.4	1.3	1.3	1.3	1.5	1.4	1.4	1.5	1.4	1.4
Precipitation	PrecAvg	1578	61	92	139	245	279	217	138	148	87	77	56	40
	PrecMax	2280	156	284	372	582	516	572	310	411	292	228	183	158
	PrecMin	929	2	9	18	92	48	61	43	16	0	0	0	2
	PrecStd	334	49	69	75	104	109	115	73	88	63	59	51	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.5	27.8	28.4	32.1	34.7	36.0	37.6	37.4	36.0	33.5	29.8	23.7
		MCWB	17.3	19.7	20.9	23.5	25.4	26.3	25.5	26.1	25.8	24.1	21.6	17.7
	2%	DB	20.7	24.9	26.3	30.3	33.0	34.6	36.2	36.0	34.4	31.3	27.2	21.0
		MCWB	15.7	18.6	19.7	22.9	24.3	26.0	25.9	26.1	25.3	22.3	19.3	15.3
	5%	DB	18.4	22.4	24.1	28.4	31.6	33.5	35.1	34.9	33.2	29.7	25.2	19.0
		MCWB	14.1	17.4	19.1	22.1	24.4	25.9	25.9	25.9	25.1	21.5	18.5	13.7
	10%	DB	16.2	19.9	22.1	26.6	29.9	32.1	33.9	33.6	31.8	28.1	23.2	17.1
		MCWB	12.7	16.4	18.3	21.7	24.0	25.8	26.0	25.8	24.6	21.0	17.6	12.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	20.7	22.9	25.1	27.1	28.1	27.9	27.7	27.1	25.4	23.0	20.0
		MCDB	21.9	25.9	26.2	29.8	31.7	32.3	33.2	33.9	32.7	30.6	26.6	22.2
	2%	WB	16.7	19.2	21.5	24.1	26.1	27.2	27.3	27.2	26.5	24.4	21.6	17.2
		MCDB	19.6	23.6	24.2	28.2	30.6	31.6	32.6	33.0	31.6	28.6	25.2	19.4
	5%	WB	14.9	18.0	20.1	23.4	25.5	26.7	26.9	26.8	26.0	23.6	20.4	15.2
		MCDB	17.7	21.6	22.7	26.9	29.5	31.0	32.2	32.1	30.9	27.0	23.2	17.6
	10%	WB	13.2	16.6	18.9	22.6	24.9	26.3	26.5	26.4	25.5	22.7	19.0	13.5
		MCDB	15.6	19.8	21.5	25.6	28.3	30.3	31.4	31.2	29.8	26.0	22.1	16.2
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.7	6.4	6.9	7.3	7.0	8.1	8.2	8.2	8.6	8.6	8.1
		MCDBR	11.8	11.8	11.3	10.4	10.6	9.6	10.4	10.6	10.5	11.4	12.5	12.5
		MCWBR	6.6	5.9	5.0	4.0	3.5	2.5	2.2	2.4	2.7	3.5	4.9	6.5
	5% WB	MCDBR	9.5	9.8	8.2	7.9	7.8	7.6	8.5	8.8	9.1	8.2	8.7	8.8
		MCWBR	5.8	5.4	4.9	4.0	3.3	2.6	2.5	2.6	2.7	3.6	4.6	5.6
Clear-Sky Solar Irradiance	taub	0.667	0.721	0.834	0.803	0.658	0.616	0.591	0.663	0.704	0.728	0.626	0.630	
	taud	1.569	1.513	1.372	1.453	1.746	1.852	1.910	1.719	1.622	1.534	1.691	1.652	
	Ebn at Noon	588	597	560	597	690	715	732	680	636	583	617	595	
	Edh at Noon	245	276	333	313	232	208	196	236	255	266	214	217	
All-Sky Solar Radiation	RadAvg	2.09	2.19	2.41	2.89	3.56	3.88	4.82	4.54	4.08	3.65	2.88	2.51	
	RadStd	0.48	0.48	0.42	0.45	0.50	0.57	0.49	0.33	0.36	0.45	0.47	0.56	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57

Nomenclature: See separate page